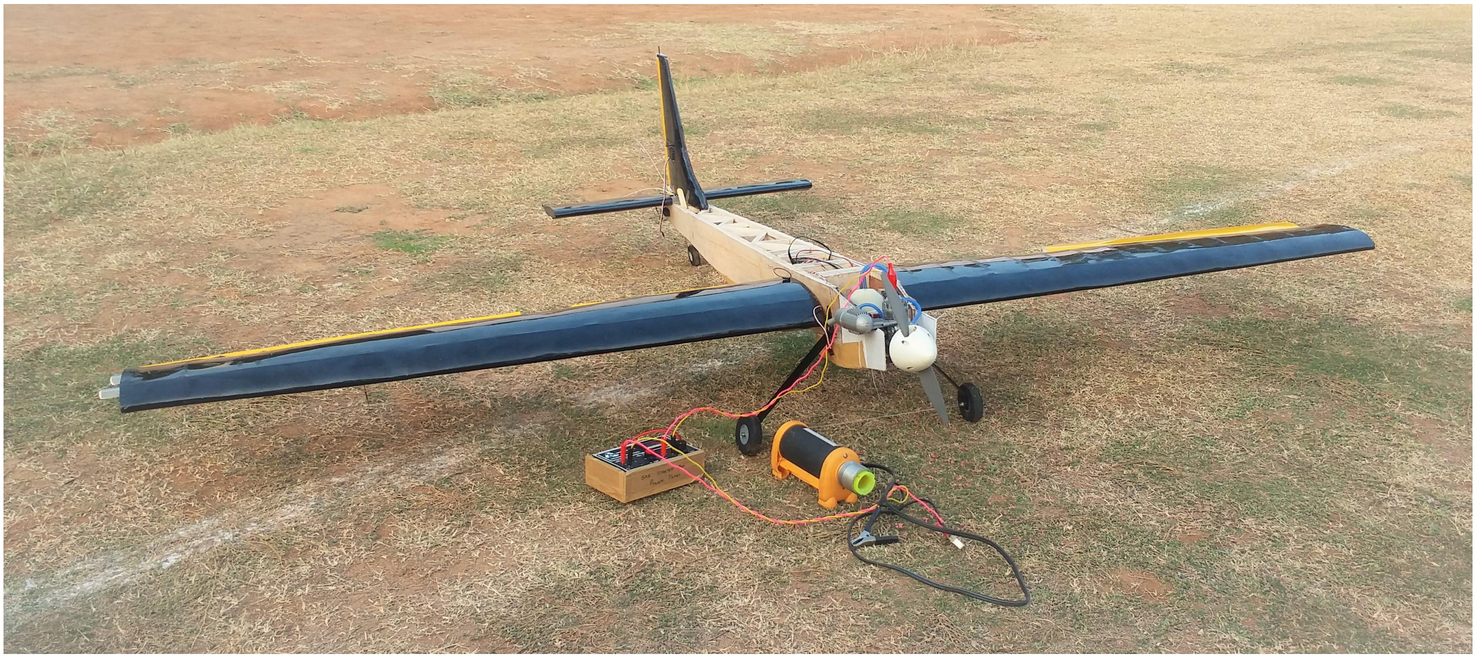
Team Nimbus(#227)
IIT Madras



Advanced Class
SAE Aero Design West 2017

Anil Kumar Pasam

G D Sunil

M Mohamed Khalid

Nikhil Kumar Gupta

Ashutosh Kumar

Perepu Sai Madhav

Rohit Jejani

APPENDIX A

SAE AERO DESIGN

STATEMENT OF COMPLIANCE

Certification of Qualification

Team Name	NIMBUS	Team Number	227
School	Indian Institute of Technology, Madras		
Faculty Advisor	Dr. H.S.N Murthy		
Faculty Advisor's Email	mhsn @ iitm.ac.in		

Statement of Compliance

As Faculty Advisor, I certify that the registered team members are enrolled in collegiate courses. This team has designed, constructed and/or modified the radio controlled aircraft they will use for the SAE Aero Design 2017 competition, without direct assistance from professional engineers, R/C model experts or pilots, or related professionals.

H.S.N. Murthy

Signature of Faculty Advisor

Team Captain Information:

Team Captain:	Nikhil Kumar Gupta
Captain's E-mail:	nikhilgupta280695@gmail.com
Captain's Phone:	(+91) 9176474258 / (+91) 9940253691

Note:

A copy of this statement needs to be included in your Design Report as page 2 (Reference Section 4.3)

Contents

Technical Design Report

1.1 Executive Summary

This document describes the journey of Team Nimbus, a team of Aerospace engineering students from Indian Institute of Technology, Madras towards participation in Society of Automotive Engineers, Aero Design West 2017 competition, Advanced class.

Our objective is to design and fabricate an aircraft that is capable of dropping payload at a remote location using a live camera feed and other sensor assists. Advanced class restricts the power plant to maximum total displacement of 0.46 cubic inches. Accordingly, we chose ASP S46 engine with the displacement of 0.46ci and a propeller of 10" diameter and 6" pitch is used based on results from testing different propellers as given in section 1.2.9.

With the objective and power plant set, we designed aircraft on the lines of *Aircraft Design: A conceptual approach* by *Daniel P Raymer* [1]. Starting from legacy data of various unmanned aerial vehicles in the same weight class, we came up with a newly designed aircraft of 8.20' wingspan and 4.79' fuselage length. The final aircraft weighs a total of 11 lbs and is projected to take a payload of 5.51 lbs.

1.2 Design Layout and Trades

1.2.1 Vehicle Configuration selection

The team decided on the mission profile based on the competition's problem statement and from experience of building UAVs.

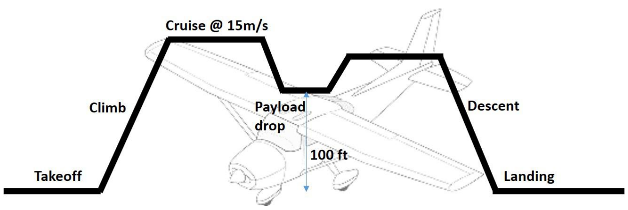


Figure 1.1: Mission Profile

Fuel weight was calculated from required endurance and engine's specific fuel consumption. Based on the desired payload capability and fuel weight required, maximum take-off weight was estimated from data fit of UAVs belonging to similar class. We estimated the take-off weight to be 16.5 lbs.

1.2.2 Airfoil selection

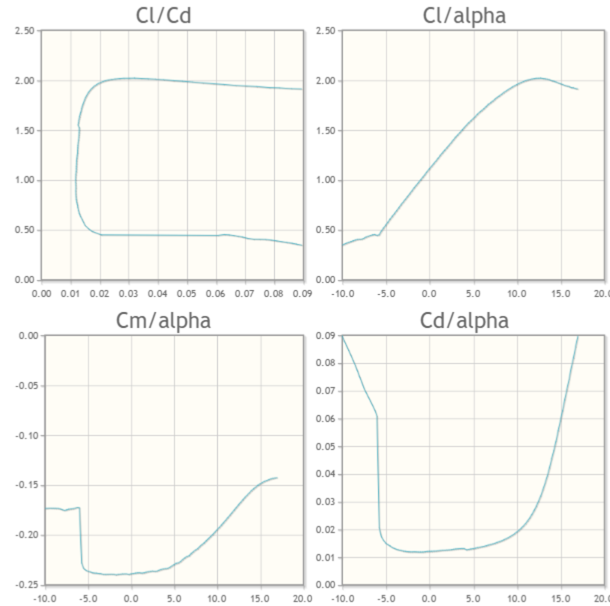
The airfoil selection was done with the intention to maximize the lift while not compromising on the stalling characteristics, drag characteristics and ease of fabrication. This was done by taking into consideration the following:

1. Lift Coefficient(C_l) v/s Angle of attack(α)
2. Drag Coefficient(C_d) v/s Angle of attack(α)
3. Drag Polar (C_d v/s C_l)
4. Pitching Moment Coefficient(C_m) v/s Angle of attack(α)

The airfoil which best satisfies these constraints is *Eppler E423*, a high-lift airfoil. Some important aerodynamic characteristics of the airfoil are summarised:

Airfoil	Eppler 423
t/c	0.125
Max. Thickness Location	0.3c
Max. Camber	0.1c
C_{L_α}	5.731
$C_{L_{\text{cruise}}}$	1.113
$C_{L_{\max}}$	2.019
L/D_{max}	68
C_L at L/D_{max}	1.212

(a) Airfoil characteristics



(b) Aerodynamic parameters of airfoil

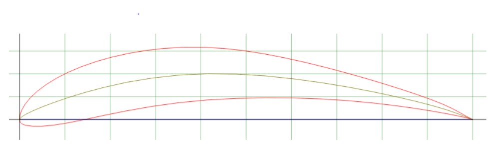


Figure 1.2: Eppler 423 airfoil

1.2.3 Wing planform design

Wing design is the one of the most critical aspect of the aircraft design process as it heavily influences the design decisions taken henceforth. Given below are some of the design decisions taken based on how it affects the aircraft's performance:

1. **Aspect Ratio**

   Choice of Aspect ratio influences maneuverability, induced drag and the bending moments acting on the aircraft. Higher the Æ, greater the stability, implying lower maneuverability. Induced drag has an inverse dependence on Æ, which means that higher aspect ratio is preferred. However, higher aspect ratio also means a longer wing, which increases the bending moment. A trade-off between these three is required for good performance. A value of around 7-10 is recommended by Raymer for UAVs. We chose AR equal to **8**. This value is chosen more towards the stability region than the maneuverability region because the mission profile doesn't require aircraft to perform maneuvers.

2. **Taper Ratio**

   Taper Ratio affects inducing drag by influencing span wise lift distribution. A taper ratio of 0.45 gives a near elliptical lift distribution, which has the least induced drag (Oswald's efficiency factor of 1). But, we went ahead with taper ratio of **0.7** for ease of fabrication.

3. **Dihedral** was not considered in the design due to manufacturing complications.

4. **Sweep** Sweep at low speed is slightly disadvantageous since it reduces the lift (airfoil faces only a component of velocity). No significant quarter chord sweep angle is chosen for aerodynamic reasons.

The wing is sized by solving the equations of motion of the aircraft at level flight condition. Level flight can be represented by the eqn $L = W = \frac{1}{2}\rho v^2 S C_L$ The design takeoff weight(W) is *16.53 lbs* and cruise velocity is *50 ft/s*. C_L for the wing can be computed from the airfoil lift coefficient, aspect ratio(AR) and taper ratio(λ) [2] as shown in $C_{L_\alpha} = \frac{C_{l_\alpha}}{1+\frac{C_{l_\alpha}}{\pi e \mathit{AR}}}$

Solving for wing area(S) in the lift equation, $S = 6.45 ft^2$ The wing dimensions were computed using wing area, aspect ratio and taper ratio. They are tabulated below.

Span	7.18'
Root Chord	1.04'
Tip Chord	0.75'
Mean Aerodynamic Chord	0.91'

Table 1.2: Wing Sizing

1.2.4 CFD analysis of the wing - winglet design

We carried out CFD analysis for the wing to make sure that lift generated by wings is sufficient to lift 16.53 lbs. For computational simplicity, flow around only half wing is simulated with a symmetry boundary condition at the root.The lengths-limits of the wing and boundary conditions of enclosure faces (refer figure 1.3). are given in table 1.3

Face	**Length (in ft)**	$\times$ **Chord**	**Boundary condition**
Front	13.12	14.5c	Velocity inlet
Back	26.24	29c	Mass outflow
Root side	0.065	0.07c	Symmetry
Tip side	13.12	14.5c	Wall
Up	13.12	14.5c	Wall
Down	13.12	14.5c	Wall

Table 1.3: Boundary lengths and conditions at various faces of enclosure

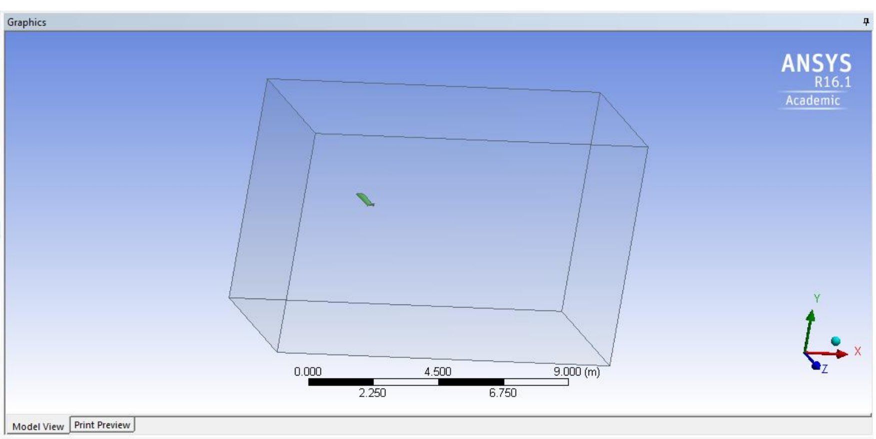


Figure 1.3: Enclosure of the wing

Characteristics

- For the faces up, down and tip side, far field pressure condition is the ideal case.

This boundary condition requires ideal gas scenario and also requires solving energy equation which increases time required for computation.

- Flow speeds being low subsonic, flow is assumed to be *incompressible.* energy equation is not solved and incompressible assumption was used.
- Due to the shortcomings in computational resources, wall boundary condition is used for the faces up, down and tip side. Since, the boundary is sufficiently far from the body, wall boundary condition is considered non-intrusive. These assumptions are found to reduce simulation time by a considerable margin and are hence used for all analyses that follow.
- Since the induced drag is major contributor to drag at low speeds when compared to parasite drag, it is analysed to a higher degree and hence *inviscid* assumption is used.
- ANSYS geometry modeller has a minimum enclosure limitation. Hence, zero distance from root could not be given. This will lead to a minor error but will affect all cases equally and hence comparison will still hold true.
- For low speed flows, pressure based solvers are found to be more efficient and hence SIMPLE scheme with default relaxation factors is used. Results from this simulation are given in figure 1.4 and also in table 1.6. It can be seen that the wing designed will be able to lift the 16.53 lbs design weight and hence the wing is finalised.

Winglet design

Along with verifying lift available, we also decided to test the efficiency of various winglets towards reduction of induced drag. We compared three winglets i.e Elliptical, Raked (similar to one used in Boeing 767-400) , Sharklet (similar to one used in Airbus A320). The geometries tested are as given in figure 1.5. The dimensions for the winglets are chosen by comparing with aircraft mentioned.

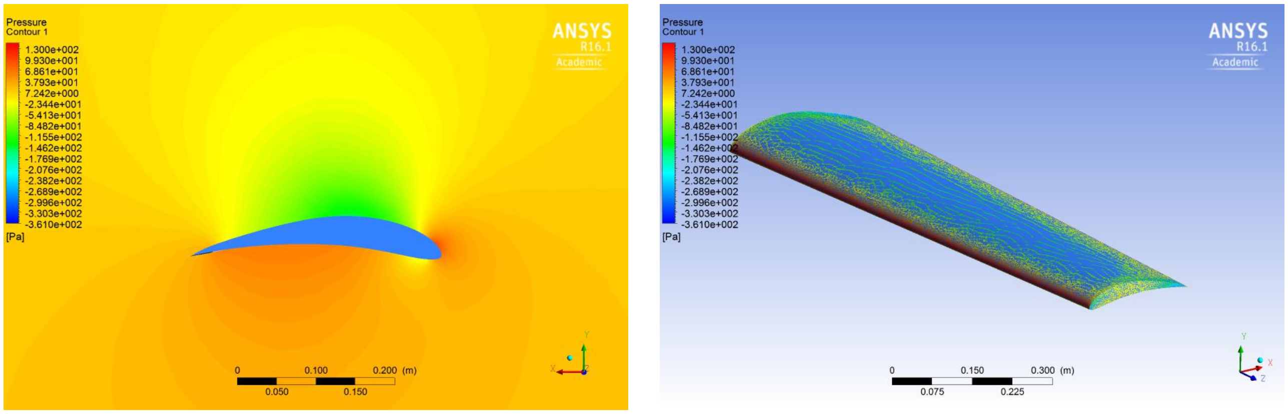


Table 1.4: Left: Chord wise pressure distribution at a distance of 1.6' from root. Right: Pressure contours accross wing surface.

Results of these calculations are given in table 1.6. From these results we concluded that the *raked winglet* is better suited at cruise conditions.

1.2.5 Fuselage Sizing

Fuselage length was calculated from the historical data, The data given in the table was used to obtain an empirical relationship between fuselage length (FL) and the take-off weight (W_0). The data was fit to a power law which in SI units is , $FL = 1.02(W_0)^{0.165}$

For a take-off weight of 16.53 lbs, the fuselage length is **4.59** ft.

1.2.6 Tail and Control Surface Sizing

The primary purpose of a tail is to provide stability as well as control to an aircraft. The tail and wing dimensions are related via the volume coefficients. *NACA 0012* was chosen as airfoil for the stabilisers.

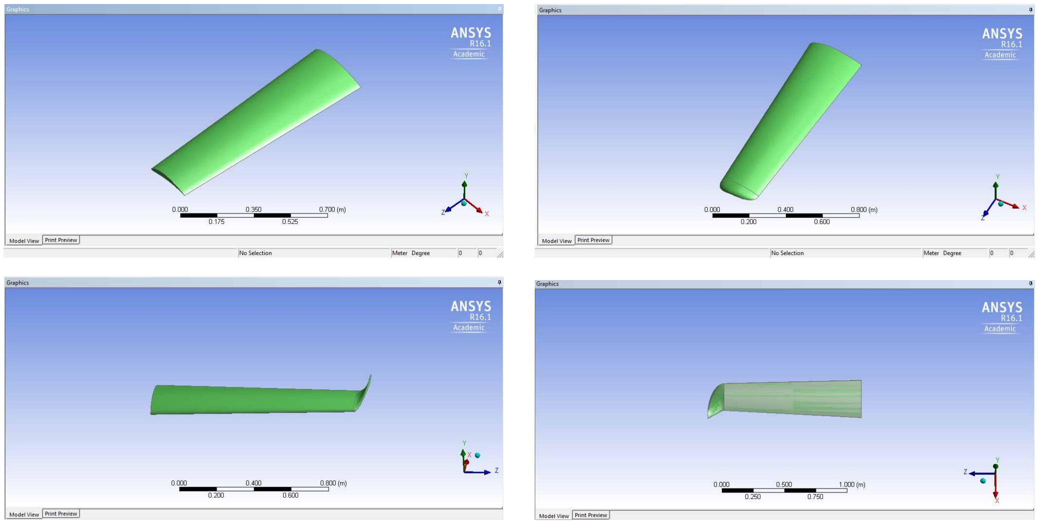


Table 1.5: The four wing configurations tested. Without winglet,Elliptical,Raked,Sharklet (from top left and clockwise)

Winglet	F_x **(Drag)**	F_y **(Lift)**	F_z **Cross force**
Without winglet	0.56 lbf	8.53 lbf	-0.08 lbf
Sharklet	0.55 lbf	9.41 lbf	-0.40 lbf
Elliptical winglet	0.56 lbf	9.05 lbf	-0.07 lbf
Raked winglet	0.55 lbf	9.5 lbf	-0.05 lbf

Table 1.6: Results from CFD analysis of various winglets

Horizontal tail

The relationship between the wing geometric parameters and the horizontal tail geometric parameters[1] is $V_{ht} = \frac{l_{ht}S_{ht}}{\bar{c_w}S_w}$. The typical value of V_{ht} is 0.6[1]. For aircraft with front-mounted propeller engine, the value of l_{ht} is around 0.6 times the fuselage length. Substituting the values of the respective parameters, $S_{ht} = 1.227ft^2$

Vertical tail

Similarly, relation between geometric parameters of wing and vertical tail, $V_{vt} = \frac{l_{vt}S_{vt}}{b_w S_w}$ The typical value of V_{vt} is 0.04[1]. Substituting the values of the respective parameters, $S_{vt} = 0.64ft^2$ The dimensions of the horizontal and vertical tail have been tabulated in 1.7,

Parameter	Horizontal tail	Vertical Tail
Aspect Ratio	3	3
Taper Ratio	1	0.3
Tail Area (S_t)	$1.55ft^2$	$0.64ft^2$
Span (b_t)	$1.916ft$	$1.11ft$
Chord at root	$0.63ft$	$1.14ft$
Chord at tip	$0.63ft$	$0.32ft$

Table 1.7: Tail sizing

Control Surface sizing

We designed the control surfaces based on legacy data and empirical relations [1].

- *Aileron* spans 50-90% of the wing-span, and is positioned towards the tip to maximize the rolling moment. Chord length of the aileron is approximately 20-25% of wing mean chord.
- *Flaps* are present towards the wing root and has chord dimensions similar to that of aileron.
- *Rudder* and *Elevator* extend from fuselage to around 90% of tail span and its chord is 25-50% of tail chord.

1.2.7 Drag analysis

Each component of the UAV contributes to parasitic drag (C_{D_0}). We calculated individual contributions of the various components - wing, fuselage, horizontal and vertical tails using component build-up method[1]. The overall parasitic drag was calculated and summarized as follows,

Component	Skin Friction Coeff.	Form Factor	Wetted Surface area	C_{Do}
Wing	0.00232	0.974	1.387	0.00392
Fuselage	0.00097	1.214	1.490	0.00219
Vertical Tail	0.00308	0.964	0.142	0.00055
Horizontal Tail	0.00309	0.964	0.279	0.00109

Table 1.8: Component-wise form-drag contribution

The induced drag co-efficient (k) is given by $k = \frac{1}{\pi e AR}$

Drag Polar : $C_D = 0.0078 + 0.0424 C_L^2$

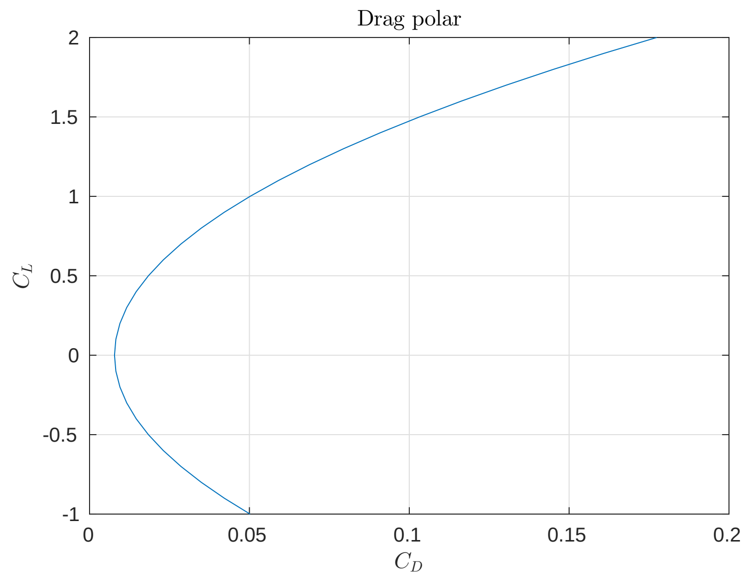


Figure 1.4: Drag Polar

1.2.8 Aircraft stability analysis

Static Margin

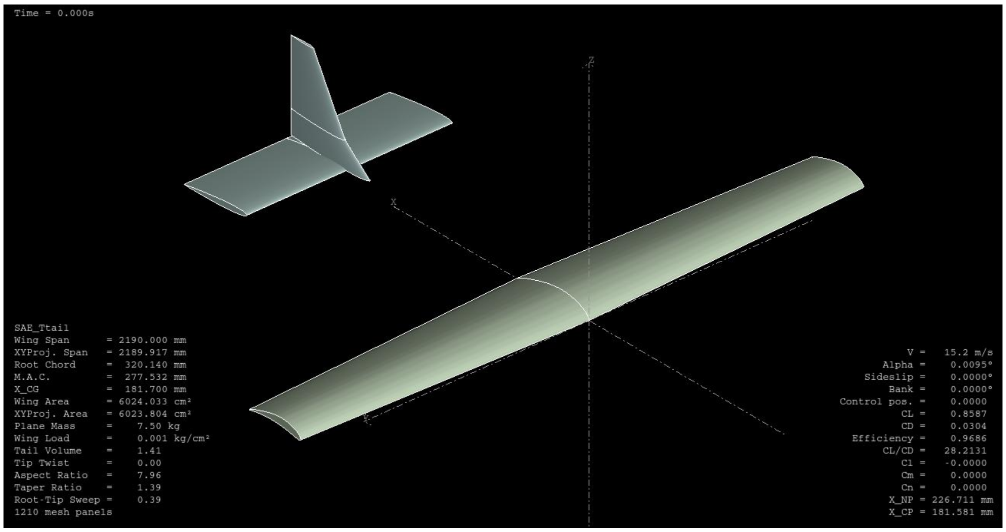


Figure 1.5: Plane definition in XFLR5

Static Margin is a measure of the of the static stability of the system. The desired value for static margin is about 5-20% [1]. Trim point with $\alpha = 0$ for level flight was used for stability analysis. The CG and Neutral Point locations were obtained from XFLR5 after mass distribution and trim data were input into it. Static Margin is **16%** in this case. Static margin should be in the desirable range even after the payload has been dropped. Static Margin after payload drop is **15.5%**. It is within the desirable limits both before and after the payload drop. So, static stability of the aircraft is guaranteed.

Dynamic Stability

XFLR5 was used for analysing static and dynamic stability of the aircraft. XFLR5 is an analysis tool for airfoils, wings and planes operating at low Reynolds Numbers. Wing, horizontal stabilizer and vertical stabilizer were defined and plane's inertia was input into the analysis tool for processing. We ran simulations for dynamic stability because static stability doesn't guarantee dynamic stability. The response to excitation for different modes are given in figures 1.6 and 1.7.

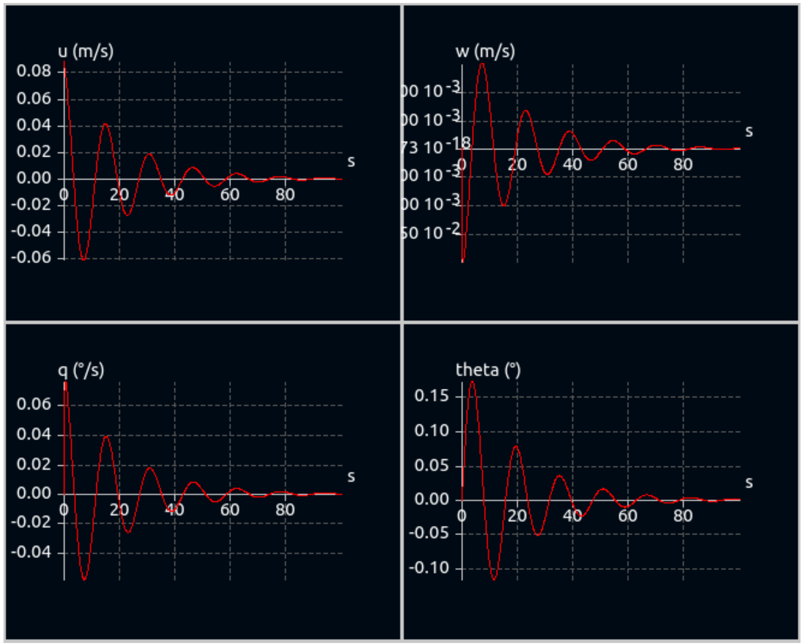


Figure 1.6: Longitudinal dynamics

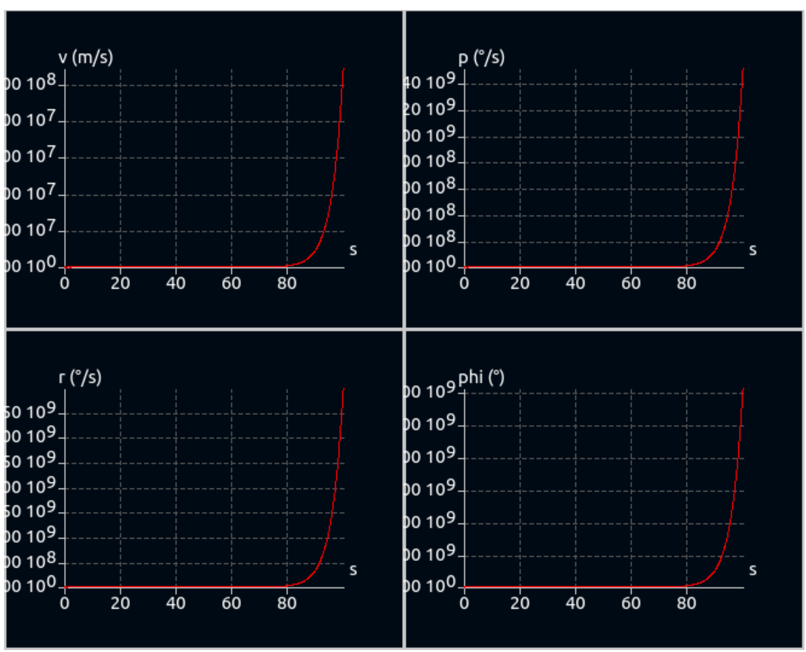


Figure 1.7: Lateral dynamics

The longitudinal modes are stable but the lateral modes aren't. Further analysis revealed that it was the spiral mode that caused this instability. To control this instability Pixhawk, an active gyro assisted flight controller is being used.

1.2.9 Propeller selection

For selection of propeller, a test bed was designed with a load cell of data acquisition frequency 1600 Hz (shown in figure 1.9). From *Top flight* propeller sizing chart given in 1.8, an engine of total displacement 0.46 ci should be using one of 10" $\times$ 5" ,10" $\times$ 6" and 11" $\times$ 7" as propeller. Since the same test bed is used for all the propellers, offsets like zero errors will affect all the propellers the same and hence the comparison will still hold true.

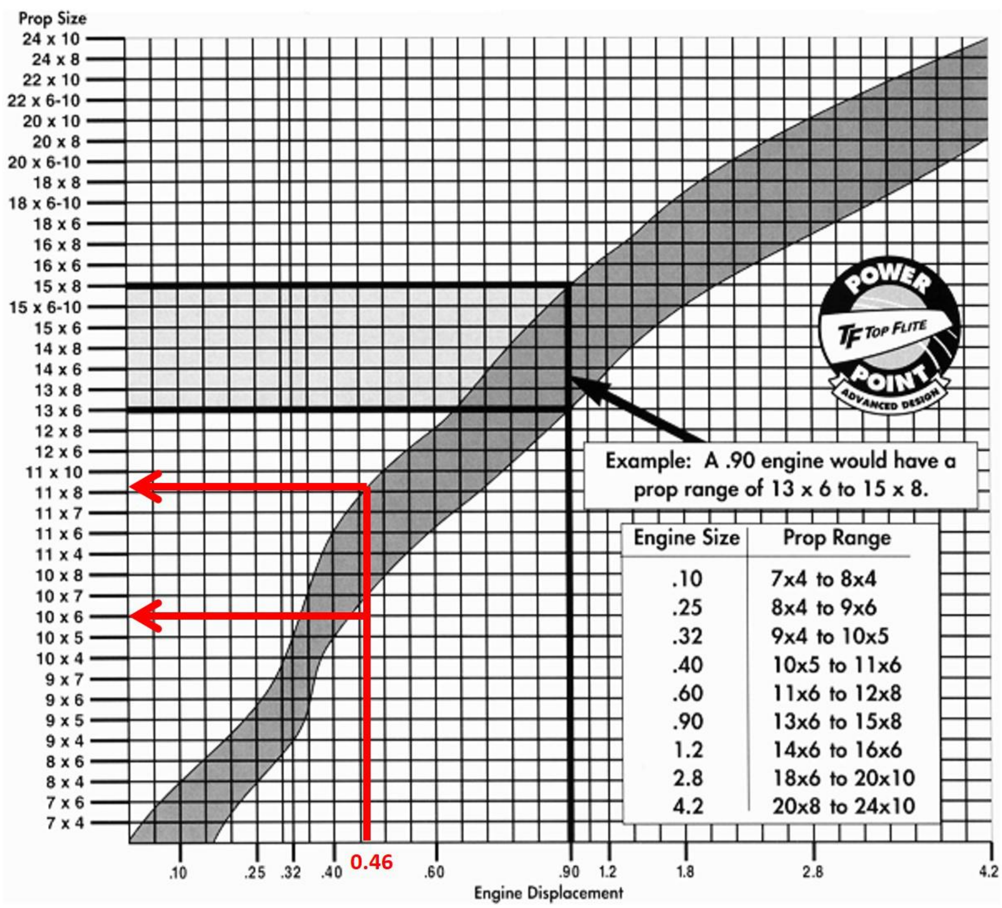


Figure 1.8: Total displacement vs Suggested propeller

We measured thrust and rpm at full throttle for the same using the test bed. Rpm that the engine produces for 10×5,10×6 and 11×7 propellers are 26800,23000 and 21000 respectively. Thrust data acquired is given in the left side of table 1.9.

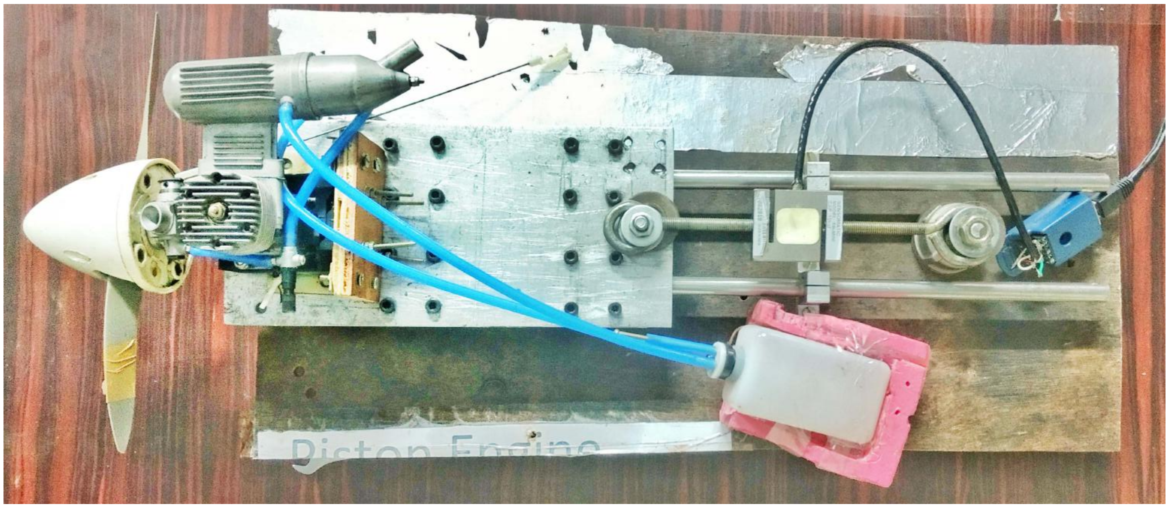


Figure 1.9: Test bed

Since this data consists of high frequency noise (likely because compression stroke produces more thrust than the expansion one), a method to get the mean of both strokes is required. The raw data is filtered using moving average method. The filtered values are given in right side of the figure 1.9.

From the data, it can be concluded that 10×6 gives the has the best max thrust of 35N(7.86 lbf). Hence, for all flight purposes, APC propeller with *10" diameter and 6" which gives the T/W ratio of 0.5 at maximum throttle and MTOW.*

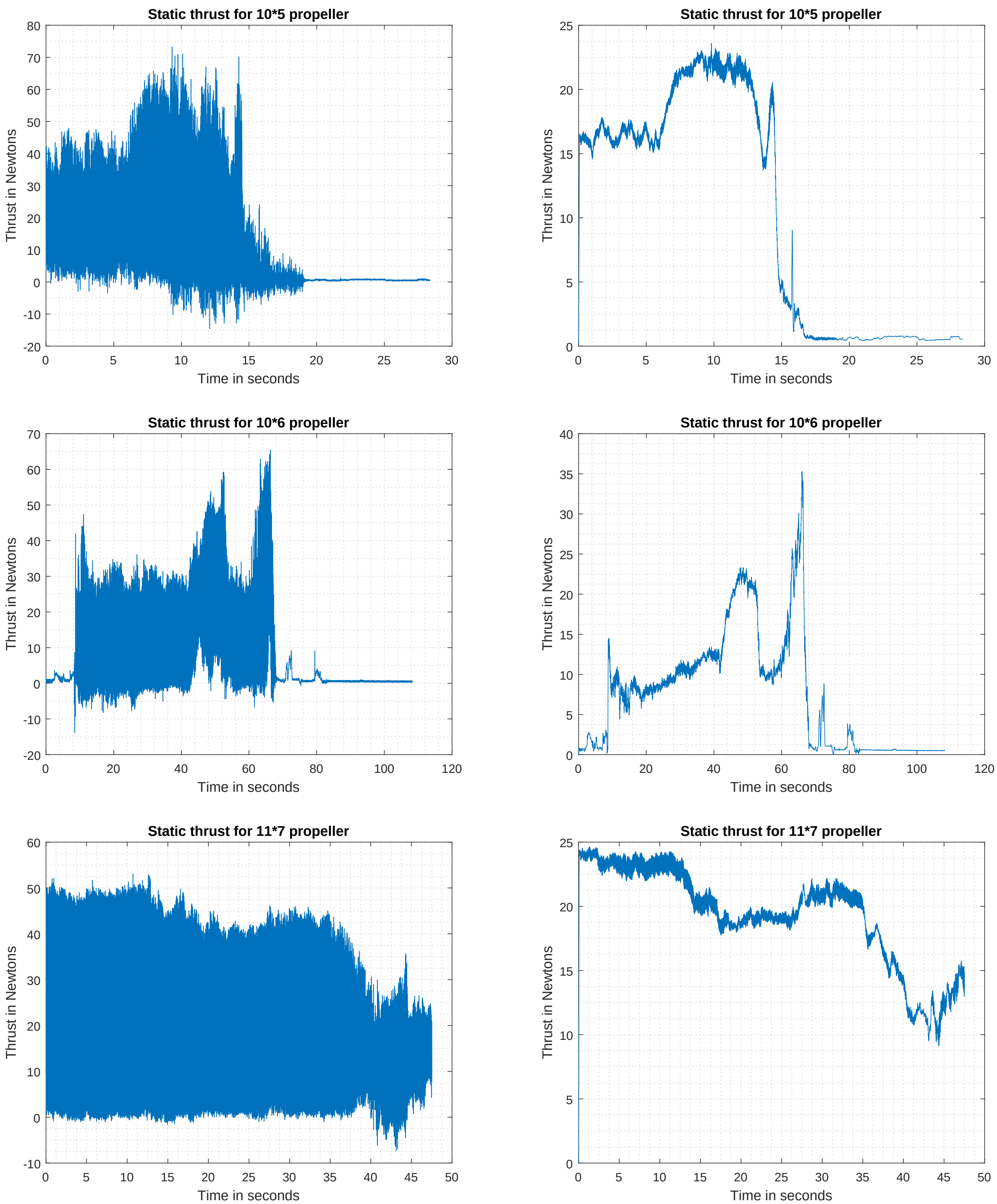


Table 1.9: Static thrust data (left: raw,right: filtered) - Top: 10" $\times$ 5", Mid: 10" $\times$ 6" and Bottom: 11" $\times$ 7"

1.3 Structural Design

1.3.1 First prototype

We aimed at making composite wings and tail in order to verify if there is a real advantage of using composites in UAVs of this weight class. From the loading analyses we concluded that Glass Fibre Reinforced Plastic (GFRP) of 0.5mm is sufficient to take the stresses that are generated. For manufacture, a foam shape of wing is first cut using in-house hot wire cutter. Resulting wing shape is shown in figure 1.10.

Figure 1.10: Foam wing

The foam body is then covered with glass fiber laminates using hand layup technique along with cavities for servo placement and also clamps to connect this wing to fuselage. The composite wing can be seen in figure 1.11 The same technique has been used to make

Figure 1.11: Composite wing

empennage section and a tail boom is used to join the fuselage (made of aluminum C sections) to empennage. Full assembly of plane is given in figure 1.12

Figure 1.12: First prototype

This plane, despite being very strong and crash proof, is found unsuitable for the competition since the structure along with essential electronics weighed 13.2 lbs which meant at most one payload can be taken.

A study of the conventional design weight breakdown is then done and it is found that conventional structure will be of lesser weight albeit being less strong (but strong enough to take 16.5 lbs weight. At this point, we shifted focus to conventional plane design i.e with ribs,spars,stringers etc.. The design of conventional structure is discussed in the subsequent subsections.

1.3.2 Second Prototype

Spar

The primary function of a wing spar is to transfer the bending stresses from the wing to the bulkhead. The wing is similar to a cantilever -maximum bending stress acts at the root section. This maximum bending stress is given by the flexure formula:

$$\sigma_{max} = \frac{M_{root} t_{max}}{2I} \tag{1.1}$$

To calculate M_{root}, we need know to the spanwise lift distribution. The lift distribution was obtained using Schrenk's method [3]. Using this method, M_{root} was found to be 485.43 lbs-inch.

Next, moment of inertia required for spar had to be estimated. Moment of inertia of spar computed using eqn (1.4) considering a load factor ($\frac{L}{W}$) of 3 and FoS of 1.5 is $7.94 \times 10^{-7}\ ft^4$. We planned to go with a three-spar design, which consists of two aluminium box channels and a carbon rod. These three spars together take the bending moment due to lift force.

Ribs

They are required to give shape to the wing and prevent buckling of the spars. It prevents buckling as the effective beam length is now the distance between two ribs and not the entire span itself. 15 ribs were used on each half-wing which also helps in ensuring that the monokote (skin) retains shape.

Stringers

Stringers were placed at definite spacing around the circumference of the ribs to make sure the shear flow in the skin doesn't lead to excessive shear stress that might lead to failure. The number of spars is decided by calculating shear flow using boom approximation.

Skin

Balsa sheets of 1.5 mm(59.05 milli inch) thickness were used wherever the curvature of wing was high making the use of monokote tough. This addition made monokote application smooth.

1.3.3 Fuselage

Bulkhead

It is the part of the fuselage that receives the loading directly from the wing. It should be highly reinforced so that it doesn't fail under the wing's loading. At the points of contact, loads due to both moment and lift force will act. This bulkhead has been intentionally over-designed for it to sustain even impact loads.

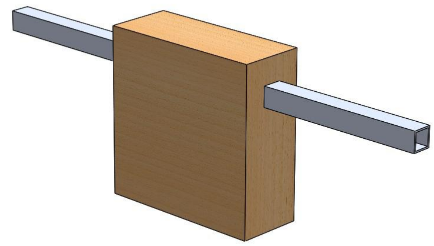


Figure 1.13: Bulkhead

Longerons

Semi-monocoque style of fuselage is in corperated because of its ease of manufacture and hence balsa longerons that transfer the load from the bulkhead to the rest of the fuselage are used. .

Engine mount

This mount experiences thrust loading from the engine and also engine vibrations. These vibrational loads need to be eliminated before the loading is transferred to the rest of the fuselage. So, vibration dampeners were placed where the engine connects to the fuselage. From a data survey, it was found that the engine mounting at an angle $3-4°$, allows a cancellation of the engine torque produced.

1.3.4 Servo sizing

All the control surfaces were actuated using 22g(0.048 lbs) servos. The sizing was done based on the maximum torque expected to be experienced by the control surface. Our aircraft is estimated to operate at around 15m/s(49.2ft/s), however a scenario of 20m/s(65.6ft/s) has been

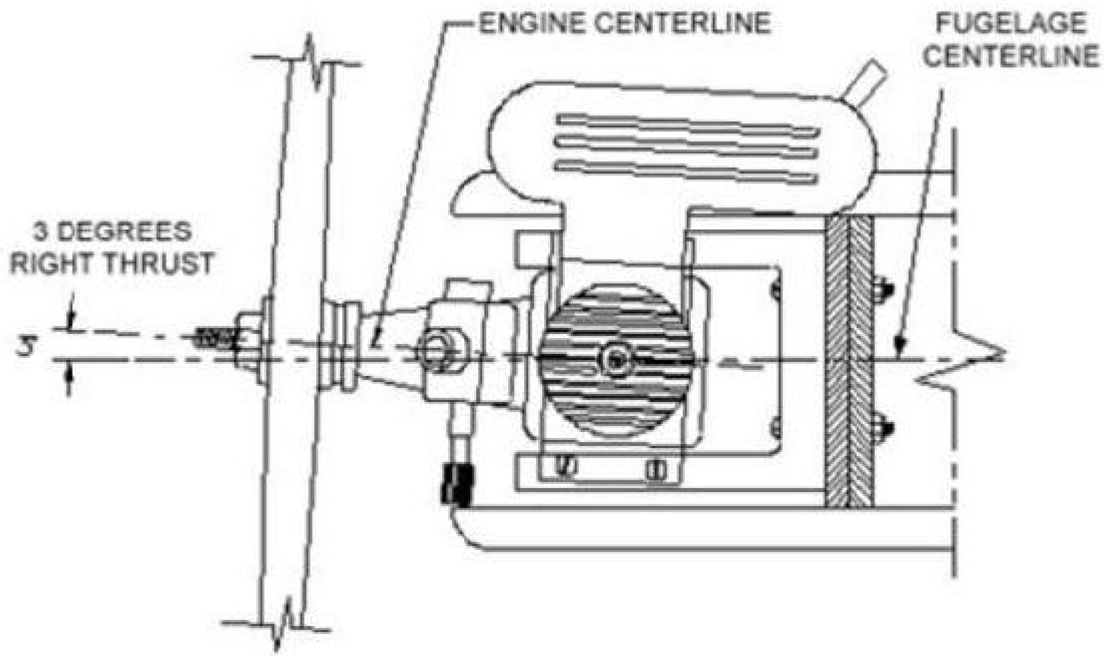
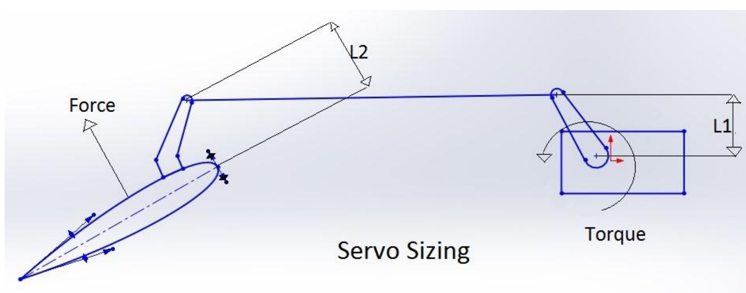


Figure 1.14: Engine mount angle

considered for nose down dive events. The control surfaces are considered as flat plates and the lift coefficient is taken to be $2\pi\alpha$ Force represents the lift on the corresponding control surface which should be balanced out by the torque provided by the servo and L_1, L_2 are the servo head and servo horn sizes respectively. Hence,



$$\text{Lift generated} \times \text{Chord} < \text{Max Servo torque}$$

Control Surface	Area (ft^2)	Max Deflection angle(deg)	Force(lbf)	Torque(lbf-inch)
Aileron	0.38	20	4.94	13.86
Flaps	0.19	30	3.37	10.39
Elevator	0.43	20	4.94	15.59
Rudder	0.22	20	2.69	7.79

Table 1.10: Servo loading

Servo size	Max output torque	Stall torque
16g Analog Servo	19.52 lbf-inch	22.07 lbf-inch
17g coreless digital metal gear	25.48 lbf-inch	29.73 lbf-inch
22g digital coreless metal gear	28.45 lbf-inch	38.35 lbf-inch
25g Metal Gear Digital cyclic Servo	16.97 lbf-inch	18.63 lbf-inch
38g Metal Gear	25.40 lbf-inch	27.17 lbf-inch

Table 1.11: Servo data

Based on the data collected, we chose Avionic 22g servo for control surfaces to include a factor of safety.

1.3.5 Landing Gear

After considering both tricycle and taildragger configurations, the latter was finalised due to various reasons including better on-ground stability,directional control, higher angle of incidence and also higher ground clearance for the propeller. Moreover, the tail wheel in this configuration is placed further away from the CG, which allows it to be light and small. In order to avoid excessive bending moments, main landing gear is attached within 10% distance from the CG.

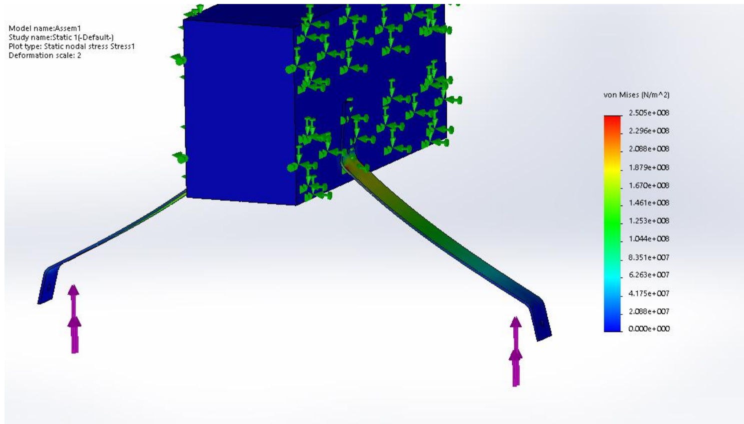


Figure 1.15: Static simulation of the landing gear

The simulation of the landing gear with a load of 7.86 lbf (>half aircraft weight) proved that it would be sufficient to take our aircraft's weight.

1.4 Analysis

1.4.1 Performance analysis

Performance characteristics were calculated using formulae available in Anderson[2].

Performance parameter	Value
Stall Velocity	34.44 ft/s
Take-off distance	46.35 ft
Max Climb Rate	16.46 ft/s
Min Turn Radius	11.71 ft

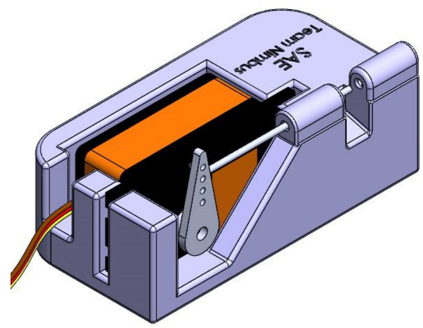
Table 1.12: Performance parameters

1.4.2 Payload

According to the problem statement, the droppable payloads have to be multiples of 2 to 2.25lbs. In addition to this, extra static payloads can be placed anywhere in the fuselage. To maximise the scoring based on the algorithm given, the optimum value of droppable and static payload were finalised.

Static (lbs)	Releasable (lbs)	Points
2.20	2.20	1.5
1.11	4.40	2
3.31	2.20	3

(a) Payload Optimisation



(b) Payload drop mechanism

Our design allowed a maximum payload of 5.511lbs, hence we decided to go with 3.51lbs static and 2.20 lbs releasable payload.
A payload mechanism was designed to allow quick drop of payload from the payload bay of fuselage.

The static payload (3.51 lbs) is placed such that the CG location coincides with the releasable payload. Hence, it was placed right behind the engine firewall. Moreover, this location moves the CG forward, thus increasing the longitudinal stability of the aircraft.

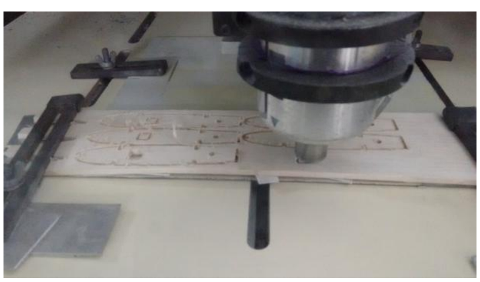
1.4.3 Electronics

Electronics consists of a 11.1V 1500mAh Lipo Battery, ESCs and the electronics bay. The electronics bay was designed for compact arrangement of all the electronic components within the aircraft. The electronic bay consists of the flight control board Pixhawk, uBlox GPS module, remote receiver, telemetry transmitter, RaspberryPi and other minor electronics. The payload drop is a semi-autonomous mechanism using computer vision assisting payload expert.

1.5 Manufacturing

The final prototype was made out of Balsa wood with plywood reinforcements. The wing, tail ribs were cut using CNC machine from 6mm(236.22 milli inch) sheets of balsa. The design for the same was done in SolidWorks and the DXF files were used to precisely make the grooves for the aluminum and carbon fiber spars.

Figure 1.16: Wing ribs assembly



(a) CNC rib cutting

(b) Ribs after cutting

The ribs with the aluminum spars were then wrapped with 1.5mm(59.05 milli inch) balsa sheets in places where skin stiffness is required and the structure is covered with monokote.

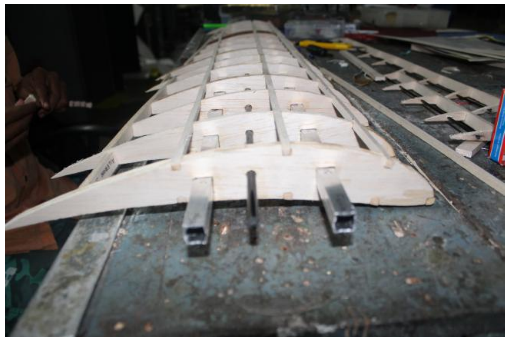


Figure 1.18: Wing after ribs assembly

The fuselage being a complicated shape, required an outer shape. For this, the surface model of the fuselage was opened up and created using corruagted plastic. Folding the sheet accordingly gave the required fuselage shape. Balsa ribs and trusses were added inside to get the required strength.

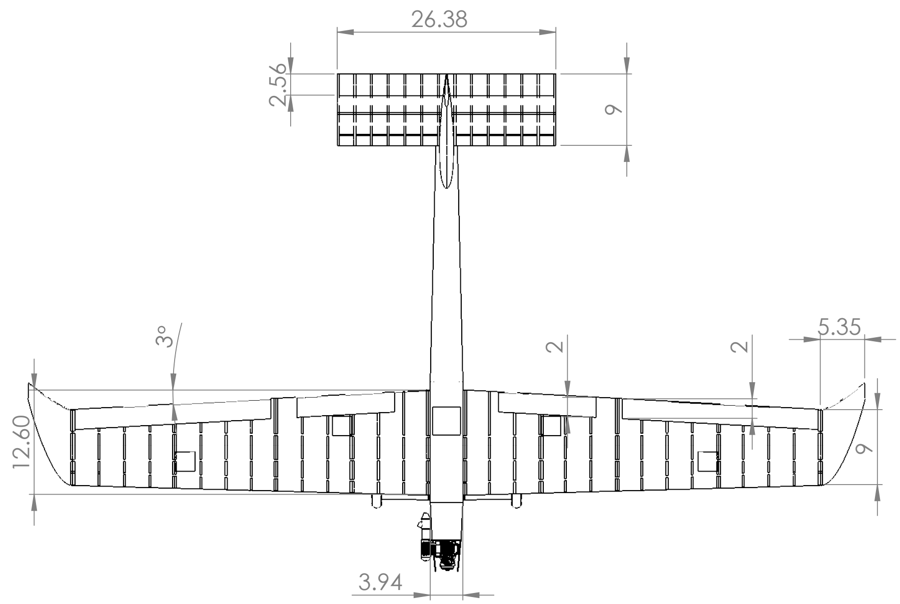
Bulkheads were carved out of Balsa blocks, through which the wing aluminum spars pass. The engine was attached to a firewall and the firewall to the fuselage.

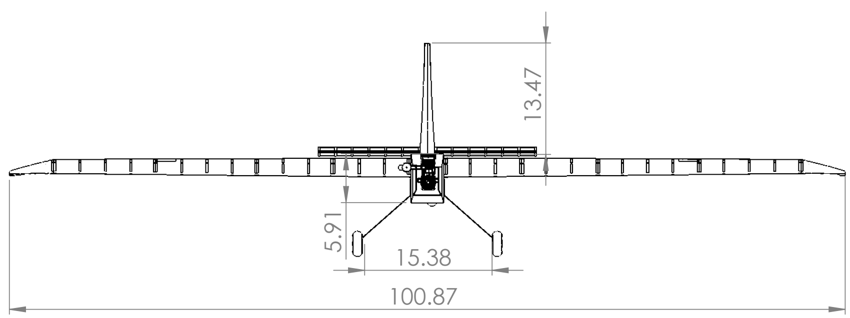
The landing gear was attached to the fuselage using ply sections to provide extra landing shock strength.

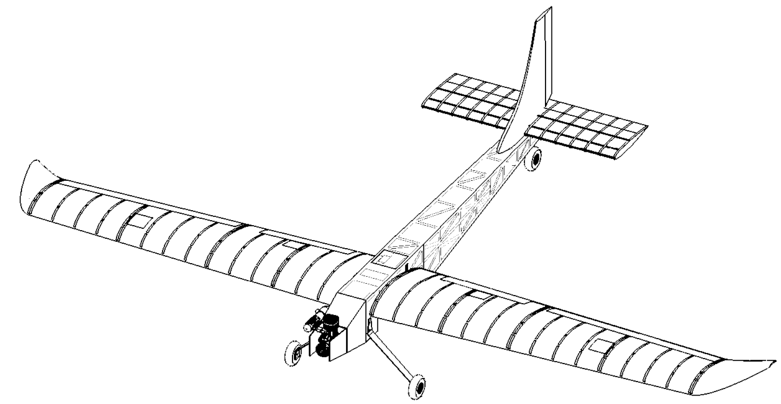
1.6 Conclusion

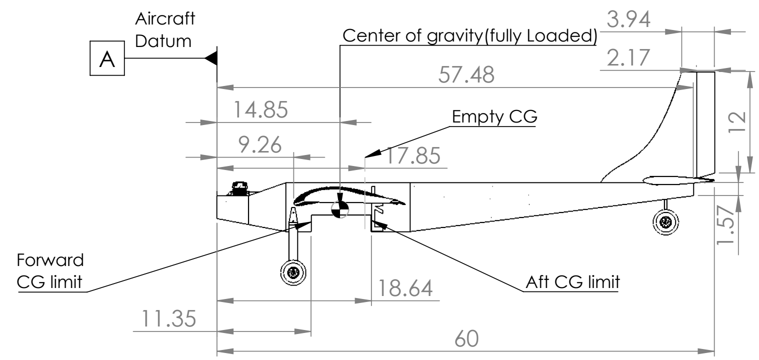
We conclude, by saying that the design of our plane is very much in accordance to satisfying the problem statement of the competition, and is ideal for lifting payloads upto 7lbs. Various calculations and simulations are in accordance to this.

Team Nimbus is very grateful to all our supporters and sponsors including Dept. of Aerospace Engineering and Center for Innovation, Indian Institute of Technology Madras.









Summary Data	
Wingspan	100.88"
Empty Weight	11.464 lbs
Engine model	ASP - 0.46 ci
Fuel Capacity	300 ml
Propeller manufacturer, dia and pitch	APC 10" x 6"
Battery	3S 1500 mAh LiPo battery

Weight and Balance information

Part Name	Location	Force (lbf)	Moment (lbf-inch)
Engine	2.5"	1.01	12.423
Fuel tank	6.0"	0.58	4.988
Payload: Static	11.1	4.50	16.65
+ Releasable	14.8"	2.5	0.000
Electronics	18.2"	1.34	4.556
Battery	19"	1.1	6.820

SAE Team Nimbus (#227)

SIZE **B**

SCALE: 1:20

Tolerance : ±1 inch

APPENDIX B

Advanced Class Tech-Data Sheet: Radio Link Budget

Supply data for each radio system used onboard aircraft. Expand number of columns as needed. Calculations for Radio control systems using 2.4GHz are optional.

Radio System Function (FPV, DAS, RC, Payload)	Units	FPV	DAS	RC	Payload Release
Operating Frequency, F	MHz	5800	915	2400	2400
Wavelength (WL) = 300 / F (MHz)	meters	0.052	0.328	0.125	0.125
Maximum Operating Range (Rng)	meters	900	500	2300	2300
Free Space Path Loss (Lfs) = 20Log(4Pi (Rng / WL))	dB	111.232	85.646	107.281	107.281
Transmitter Brand/Model		TS351	3DR Telemetry	Flysky FS-i6	AvionicsRCB 6i
Transmitter Power (Pt)	dBm	22	20	<20	<20
Number of Transmitter Channels	---	8	1	6	6
Transmit Antenna Gain (Gt)	dB	3	2		
Transmit Antenna Polarization (H, V, RHC, LHC)		V	H / V		
Transmit Losses (Lt)	dB	-	-		
Receive Losses (Lr)	dB	-	-		
Receive Antenna Gain (Gr)	dB	11	12		
Transmit Antenna Polarization (H, V, RHC, LHC)		V	H / V		
Processing Gain (Gpr)	dB	-	-		
Polarization Loss (Lpol)	dB	-	-		
Power Rcvd (Pr) = Pt + Gt + Gr + Gpr – Lfs – Lt – Lr – Lpol	dBm	46.98	-		
Receiver Signal Power Required Pmin.	dBm	36	-		
Signal Margin = Pr – Pmin	dB	10.98	-		

FPV Transmitter Operating Frequencies Available (MHz).

If your video transmitter or data-link is set to a fixed frequency which can be changed with a switch, please list the available frequencies to aid the event in reducing frequency conflicts. If your system is spread spectrum, list the range of frequencies on which it can operate.

5645	5885						
5665	5905						
5685	5925						
5705	5945						

Reference: https://en.Wikipedia.org/wiki/Link_budget

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